ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 1 · PRACTICE C · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Robotics and Industrial Automation paper 1 · practice c.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 D ✓ 1.1.3 A ✓ 1.1.4 C ✓ 1.1.5 A ✓ 1.1.6 B ✓ 1.1.7 D ✓ 1.1.8 C ✓ 1.1.9 A ✓ 1.1.10 B ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 False ✓ 1.2.4 False ✓ 1.2.5 True ✓ (5)

1.3 1.3.1 teach pendant ✓ 1.3.2 slicer ✓ 1.3.3 latency ✓ 1.3.4 LiDAR ✓ 1.3.5 mesh network ✓ (5)

[20]

TOTAL SECTION A: [20]

SECTION B

QUESTION 2

2.1 Electric: accurate, clean and easy to control ✓ / limited force for its size, needs gearboxes ✓. Hydraulic: very large forces, holds loads well ✓ / oil leaks, noisy, needs a power pack ✓. Pneumatic: fast, cheap, clean ✓ / air is compressible, so accurate positioning is difficult ✓. (6)

2.2 Point-to-point: the robot moves between taught points and the path between them is not controlled (e.g. spot welding, pick and place) ✓. Continuous path: the whole path is controlled, so the tool follows an exact line or curve (e.g. arc welding, painting, gluing) ✓. (2)

2.3 Any sensible examples ✓✓✓, e.g. 2.3.1 lights, alarms or geysers controlled from a phone; 2.3.2 soil-moisture sensors that switch irrigation on, livestock trackers; 2.3.3 wearable heart-rate monitors that send data to a doctor. (3)

2.4 Any ONE risk with its solution ✓✓, e.g. hackers taking control of devices or stealing data – use strong passwords, encryption, firmware updates and a separate network for IoT devices. (2)

[13]

QUESTION 3

3.1 In order ✓✓✓✓: design the finger in CAD; export it as an STL file; slice it (set layer height, infill, supports) to produce G-code; print it; remove supports and finish (sand, fit soft pads), then test the fit. (4)

3.2 A higher infill percentage makes the part stronger and heavier ✓ but takes longer and uses more filament ✓. Thinner layers give a smoother surface and finer detail ✓ but increase the printing time; thicker layers print faster with a rougher finish ✓. (4)

3.3 Printed parts are never exactly the designed size (plastic shrinks and spreads slightly) ✓; a small clearance makes sure holes and mating parts still fit together without forcing or being too loose ✓. (2)

3.4 Any THREE with properties ✓✓✓: PLA – easy to print, rigid, low temperature; ABS – tough, heat resistant, needs a heated bed / enclosure; PETG – strong, slightly flexible, food safe; TPU – flexible, rubbery (good for soft grip pads); nylon – very strong and wear resistant. (3)

[13]

QUESTION 4

4.1 Star: every device is connected to a central switch ✓✓; advantage – one faulty cable affects only one device / easy to add devices ✓. Bus: all devices share one main cable (backbone) ✓✓; advantage – uses little cable, cheap and simple ✓. (6)

4.2.1 The amount of data that can be sent over a network connection in a given time ✓, measured in bits per second (e.g. Mbps) ✓. (2)

4.2.2 A unique numerical address given to every device on a network ✓ so that data can be sent to and from the right device ✓. (2)

4.3 Any TWO fieldbus ✓✓: Profibus, Profinet, Modbus, DeviceNet, CANopen, EtherNet/IP, AS-i. Any TWO wireless ✓✓: Wi-Fi, Bluetooth, ZigBee, cellular (4G / 5G), LoRaWAN. (4)

4.4 A switch connects devices within the same local network and sends data to the right device using MAC addresses ✓; a router connects different networks (e.g. the plant network to the internet) and directs data using IP addresses ✓. (2)

[16]

QUESTION 5

5.1.1 Six axes ✓ (1)

5.1.2 Reach: the maximum distance from the robot's base to the wrist / tool that the arm can extend to ✓✓. Repeatability: how closely the robot returns to the same taught point again and again (e.g. ± 0,05 mm) ✓✓. (4)

5.2 $F = m \times g = 12 \times 9{,}8$ ✓ $= 117{,}6$ N ✓; $\tau = F \times r$ ✓ $= 117{,}6 \times 0{,}4$ ✓ $= 47{,}04$ N·m ✓ (5)

5.3 $W = 5 \times 9{,}8 = 49$ N ✓; $F = W \div (\mu \times n)$ ✓ $= 49 \div (0{,}5 \times 2)$ ✓ $= 49$ N per finger ✓ (4)

5.4 Any THREE ✓✓✓: assembling small electronic parts; pick and place; packaging; screwing and inserting parts; dispensing glue; laboratory sample handling. (3)

[17]

QUESTION 6

6.1 With the right hand, the thumb points along +X, the index finger along +Y and the middle finger along +Z ✓; this gives a standard, consistent direction for all three axes of the robot's frame ✓. (2)

6.2 The analogue sensor gives a varying voltage that the microcontroller's ADC changes into a number (e.g. 0–1023) ✓✓; the program uses the value to decide the speed and sends a PWM signal through a motor driver, where a larger duty cycle gives a higher average voltage and speed ✓✓. (4)

6.3 A mechanical switch's contacts bounce, giving several quick on-off signals when it is pressed once ✓; debouncing (a short delay or a capacitor) makes the microcontroller read only one clean press ✓. (2)

6.4 The microcontroller can supply only a few milliamps at 5 V ✓; the relay lets this small signal switch the 230 V lamp safely, keeping the high-voltage circuit electrically isolated from the microcontroller ✓. (2)

6.5 The TCP is the working point at the tip of the tool (e.g. the centre between gripper fingers or the tip of a welding torch) ✓; all positions and movements are calculated for this point ✓, so if it is set wrongly the tool misses its target or collides ✓. (3)

[13]

QUESTION 7

7.1 Any TWO ✓✓: breadboard view; schematic (circuit diagram) view; PCB view. (2)

7.2 It shows the logic and order of the steps and decisions before coding ✓, so errors are found early and the program is easier to write and debug ✓. (2)

7.3 Any FOUR ✓✓✓✓: does it perform the required function; accuracy / repeatability; speed or cycle time; safety; reliability over many cycles; power consumption / battery life; cost; ease of use. (4)

[8]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE